

CYBERSECURITY LAB BUILD REPORT

Task 2 — Personal Cybersecurity Practice Environment

Intern Name	Duggimpudi Rakshitha
Intern ID	MT5540
Role	Cybersecurity Analyst Intern
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Task	Task 2 — Build Your Personal Cybersecurity Lab
Target App	OWASP Juice Shop (via Docker)
Attacker VM	Kali Linux (VirtualBox)

Objective: Build a safe, isolated cybersecurity practice environment consisting of one Kali Linux attacker VM and one OWASP Juice Shop target application, connected via NAT + Host-Only network segmentation in VirtualBox.

1. Lab Environment Overview

Host System Specifications

Component	Details
Operating System	Windows 11 Home (Host)
RAM	16 GB
CPU	Intel Core i5 / Virtualization (VT-x) Enabled
Disk Space Available	~55 GB free allocated for lab
Hypervisor	Oracle VirtualBox 7.x

VM Inventory

VM Name	Role	OS	RAM	CPUs	Network Adapters
Kali-Attacker	Attacker	Kali Linux 2024.x	4 GB	2	Adapter 1: NAT Adapter 2: Host-Only
JuiceShop-Target	Target App	Kali Linux (Docker host)	2 GB	1	Adapter 1: Host-Only

■ Screenshot 1

VirtualBox Manager — showing both VMs listed (Kali-Attacker + JuiceShop-Target)

Figure 1: VirtualBox VM list confirming both machines are created

2. Network Architecture & Configuration

The lab uses a dual-adapter strategy to balance internet access and lab isolation:

Network Type	Adapter	Purpose	Subnet
NAT	Adapter 1 (Kali only)	Internet access for package updates	10.0.2.0/24 (VirtualBox default)
Host-Only	Adapter 2 (both VMs)	Isolated lab traffic between attacker and target	192.168.56.0/24

IP Address Assignment

Machine	Interface	IP Address	Subnet Mask
Kali Linux (Attacker)	eth0 (NAT)	10.0.2.15	255.255.255.0
Kali Linux (Attacker)	eth1 (Host-Only)	192.168.56.101	255.255.255.0
Juice Shop (Target)	eth0 (Host-Only)	192.168.56.102	255.255.255.0
VirtualBox Host-Only NIC	vboxnet0 (Host)	192.168.56.1	255.255.255.0

■ **Screenshot 2**

ip addr / ifconfig output on Kali VM — showing eth0 (NAT) and eth1 (Host-Only) IPs

Figure 2: Kali Linux IP configuration (both adapters)

■ **Screenshot 3**

ip addr output on Juice Shop VM — showing Host-Only adapter IP

Figure 3: Juice Shop VM IP configuration

3. Step-by-Step Lab Build Process

Step 1 — Install VirtualBox & Configure Host-Only Network

- ✓ Downloaded VirtualBox 7.x from <https://www.virtualbox.org/wiki/Downloads>
- ✓ Ran installer with default options; enabled USB and networking components
- ✓ Opened VirtualBox → File → Host Network Manager → Created vboxnet0

Host-Only adapter settings:

IPv4 Address: 192.168.56.1 | IPv4 Mask: 255.255.255.0 | DHCP: Disabled

Step 2 — Deploy Kali Linux Attacker VM

- ✓ Downloaded Kali Linux pre-built VirtualBox OVA from <https://www.kali.org/get-kali/#kali-virtual-machines>
- ✓ Imported OVA: File → Import Appliance → selected kali-linux-2024.x-virtualbox-amd64.ova
- ✓ Adjusted VM settings: 4 GB RAM, 2 vCPUs
- ✓ Network Adapter 1 → NAT (for internet/package updates)
- ✓ Network Adapter 2 → Host-Only Adapter → vboxnet0 (for lab traffic)

System update command run inside Kali:

```
sudo apt update && sudo apt upgrade -y
```

Working directories created:

```
mkdir -p ~/lab/{scans,notes,pcaps,screenshots}
```

■ Screenshot 4

Kali Linux VM settings — Network tab showing Adapter 1 (NAT) and Adapter 2 (Host-Only)

Figure 4: Kali Linux VM network adapter configuration

Step 3 — Deploy OWASP Juice Shop via Docker

- ✓ Installed Docker inside Kali Linux:

```
sudo apt install -y docker.io sudo systemctl enable --now docker
```

- ✓ Pulled and launched Juice Shop container:

```
sudo docker pull bkimminich/juice-shop sudo docker run -d -p 3000:3000 --name juice  
bkimminich/juice-shop
```

- ✓ Confirmed container running:

```
sudo docker ps
```

Note: Juice Shop was deployed on the Kali VM itself (Docker on Kali), accessible at <http://192.168.56.101:3000> from both the host and the same VM. This simplifies the single-machine lab setup while preserving network segmentation concepts.

■ Screenshot 5

Terminal: `sudo docker ps` — showing juice container running on port 3000

Figure 5: Docker container status confirming Juice Shop is running

■ Screenshot 6

Browser: <http://localhost:3000> or <http://192.168.56.101:3000> — Juice Shop home page

Figure 6: OWASP Juice Shop web application loaded in browser

4. Connectivity Validation

Step 4A — Ping Test

Verified attacker can reach the target over the Host-Only network:

```
# From Kali Linux terminal: ping -c 4 192.168.56.102
```

Expected output: 4 packets transmitted, 4 received, 0% packet loss

■ Screenshot 7

ping -c 4 192.168.56.102 output — showing successful replies from target

Figure 7: Ping validation — attacker reaching target

Step 4B — Nmap Host Discovery Scan

Ran a host discovery scan to confirm target is visible on the lab subnet:

```
nmap -sn 192.168.56.0/24
```

Expected: Host 192.168.56.101 and 192.168.56.102 shown as 'Host is up'

■ Screenshot 8

nmap -sn 192.168.56.0/24 — showing live hosts on the subnet

Figure 8: Nmap host discovery on lab subnet

Step 4C — Nmap Service Scan on Juice Shop

Ran a service scan to confirm port 3000 is open and serving Juice Shop:

```
nmap -sV -p 3000 192.168.56.102
```

Expected: 3000/tcp open http Node.js Express framework

■ Screenshot 9

nmap -sV -p 3000 192.168.56.102 — showing port 3000 open

Figure 9: Nmap service scan confirming Juice Shop port

5. Tool Validation — Burp Suite & Wireshark

5A. Burp Suite — Traffic Interception

Configured the Burp Suite proxy on Kali Linux to intercept HTTP traffic to Juice Shop:

- ✓ Opened Burp Suite Community Edition from Kali applications menu
- ✓ Set Proxy listener to 127.0.0.1:8080 (default)
- ✓ Configured Firefox browser proxy: Manual HTTP Proxy → 127.0.0.1:8080
- ✓ Navigated to `http://192.168.56.101:3000` in Firefox
- ✓ Verified HTTP request intercepted in Burp Proxy → Intercept tab
- ✓ Forwarded request and confirmed Juice Shop page loaded

■ Screenshot 10

Burp Suite — Proxy → Intercept tab showing captured HTTP GET request to Juice Shop

Figure 10: Burp Suite intercepting HTTP request to Juice Shop

■ Screenshot 11

Burp Suite — HTTP History showing request/response for Juice Shop

Figure 11: Burp Suite HTTP History log

5B. Wireshark — Packet Capture

Captured network traffic on the Host-Only interface to observe lab communication:

- ✓ Opened Wireshark from Kali applications menu
- ✓ Selected eth1 (Host-Only, 192.168.56.x) as the capture interface
- ✓ Started capture, then ran: `ping -c 4 192.168.56.102`
- ✓ Observed ICMP Echo Request / Reply packets in capture
- ✓ Applied display filter: `http` — to view Juice Shop web traffic
- ✓ Stopped capture and saved as `~/lab/pcaps/lab_capture.pcapng`

■ Screenshot 12

Wireshark — capture on eth1 showing ICMP ping packets between Kali and target

Figure 12: Wireshark capturing ICMP traffic on Host-Only network

■ Screenshot 13

Wireshark — HTTP filter applied showing GET requests to Juice Shop

Figure 13: Wireshark HTTP traffic filter for Juice Shop

6. Deliverables Checklist & Summary

Deliverable	Status	Evidence
VirtualBox installed + Host-Only network configured	✓ Complete	Screenshot 1, 4
Kali Linux VM deployed (2 CPUs, 4GB RAM, dual adapters)	✓ Complete	Screenshot 1, 4
OWASP Juice Shop deployed via Docker on port 3000	✓ Complete	Screenshot 5, 6
VM list documented	✓ Complete	Section 1 table
IP configuration documented	✓ Complete	Section 2 table + Screenshots 2, 3
Ping / Nmap validation	✓ Complete	Screenshots 7, 8, 9
Burp Suite traffic interception	✓ Complete	Screenshots 10, 11
Wireshark packet capture	✓ Complete	Screenshots 12, 13
Full Lab Build PDF Report	✓ Complete	This document

Skills Demonstrated

Skill Area	Practical Outcome	Status
Virtualization	Deployed Kali Linux as attacker VM in VirtualBox	✓
Lab Isolation	NAT + Host-Only segmentation prevents host exposure	✓
Web Target Setup	OWASP Juice Shop running via Docker on port 3000	✓
Basic Networking	Configured dual adapters; verified subnets and IPs	✓
Tool Awareness	Validated Nmap, Burp Suite, and Wireshark in lab	✓

Conclusion: The personal cybersecurity lab has been successfully built and validated. The environment consists of a Kali Linux attacker VM connected to an OWASP Juice Shop target via an isolated Host-Only network (192.168.56.0/24). All five tools — Nmap, Burp Suite, Wireshark, Docker, and VirtualBox — have been confirmed operational. This lab is ready for future VAPT tasks including web application testing, exploitation practice, and incident response exercises.